Supplementary Material

Reliability Optimization of Weighted k-out-of-n:G Systems under Dual Fuzziness

Kadir Sarıkaya

This document collects material that supports the main article without being needed to follow its primary argument: a mechanical trace of the dual-fuzzy evaluation rule, the static reliability and both phases of the exact search (Section S1); the script-level provenance of every reported result, the verification checks and the computational environment (Section S2); the verification record of the uncertainty-theory oracle (Section S3); the Pharmaceutical-derived instance with the uncertainty-theory comparison (Section S4); the Fyffe-derived instance (Section S5); further results on the IEEE RTS-79 case study (Section S6); the details of the model baselines (Section S7); the replication on the two derived instances (Section S8); the scaling study, the ablation and the genetic-algorithm comparison (Section S9); the proofs (Section S10); and the notation (Section S11). Section numbers here are prefixed S and are cited from the main text as S1–S11.

S1 Illustrative Example: Exact Calculation Trace

This section traces the evaluation rule and the static reliability on a three-component system. Consider three components with reliabilities $p_1 = 0.95$, $p_2 = 0.90$ and $p_3 = 0.85$. Their functional capacities (weights) are modeled directly as Triangular Fuzzy Numbers (TFNs), defined via their lower, modal, and upper bounds $\tilde{w}_i = (w^0_{i,L}, w_i, w^0_{i,U})$:

- $\tilde{w}_1 = (4.32, 4.80, 5.28)$
- $\tilde{w}_2 = (4.50, 5.00, 5.50)$
- $\tilde{w}_3 = (4.50, 5.00, 5.50)$

Simultaneously, the system's operational target is defined by an expert-elicited fuzzy baseline threshold:

$$\tilde{k} = (6.0, 8.0, 10.0) \tag{S.1}$$

To evaluate the success condition ($\tilde{W}_S \tilde{\geq} \tilde{k}$), we apply the Hurwicz evaluation rule. Under the pessimistic evaluation stance ($\gamma = 0$), which by Proposition 1 corresponds to the necessity ranking criterion of Dubois and Prade (1983) and Inuiguchi and Ramík (2000), a subset is accepted only if its lower capacity endpoint is at least the upper endpoint of the threshold, $W^{\alpha}_{S,L} \geq k^{\alpha}_U$.

Assume a configuration that engages all three modules ($\mathbf{x} = [1, 1, 1]$). We trace its static reliability at the cut level $\alpha = 0.5$.

1. Evaluating Component Capacities and Subset Weight

First, we extract the $\alpha = 0.5$ lower-bound capacities for the active components using the algebraic interval transformation $w^{\alpha}_{i,L} = w^0_{i,L} + \alpha(w_i - w^0_{i,L})$:

- $w^{0.5}_{1,L} = 4.32 + 0.5(4.80 - 4.32) = 4.56$
- $w^{0.5}_{2,L} = 4.50 + 0.5(5.00 - 4.50) = 4.75$
- $w^{0.5}_{3,L} = 4.50 + 0.5(5.00 - 4.50) = 4.75$

When all three components are operational (forming subset $S = \{1, 2, 3\}$), the aggregated pessimistic boundary evaluates to $W_{S,L}^{0.5} = 4.56 + 4.75 + 4.75 = 14.06$.

2. Extracting Threshold and Checking Subset Feasibility

Next, we extract the upper endpoint of the threshold:

$$k_U^{0.5} = 10.0 - 0.5(10.0 - 8.0) = 9.0 \tag{S.2}$$

Evaluating the core success condition ($W_{S,L}^{\alpha} \geq k_U^{\alpha}$) yields:

$$14.06 \geq 9.0 \quad \textbf{(Condition Satisfied)} \tag{S.3}$$

The full set $S = \{1, 2, 3\}$ is therefore accepted at $\alpha = 0.5$.

3. Determining Functional Subsets

We evaluate if the system survives the failure of individual modules. If component 1 fails, the remaining aggregated capacity is $W_{S',L}^{0.5} = 4.75 + 4.75 = 9.5$. Since $9.5 \geq 9.0$, the system remains operational. Similarly, the removal of component 2 or component 3 yields remaining weights of $9.31 \geq 9.0$. Thus, all subsets with at least two components ($S \in \{\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$) are validated success states.

4. Calculating the Static Reliability

The static reliability $R_{\text{sys}}^{0.5,0}$ of Equation (11) of the main article sums the probabilities of the four mutually exclusive success configurations found in Step 3:

$$\begin{aligned} R_{\text{sys}}^{0.5,0} &= p_1p_2p_3 + p_1p_2(1 - p_3) + p_1p_3(1 - p_2) + p_2p_3(1 - p_1) \\ &= 0.72675 + 0.12825 + 0.08075 + 0.03825 = 0.974. \end{aligned} \tag{S.4}$$

Conclusion of the Trace

The α-cut endpoints of the capacities and of the threshold determine which subsets are accepted, and the static reliability (0.974) follows from the probabilities of the accepted subsets.

S1.1 Optimization Trace: The Two-Phase Exact Search

We trace Algorithm 1 of the main article on the three components above, now read as three subsystems with catalogs, in the cell $(\alpha, \gamma) = (0.5, 0)$, which by Proposition 1 is the necessity constraint $N(\tilde{W}_S \geq \tilde{k}) \geq 0.5$. Subsystem 1 offers two options: option 1, the unit above ($c_{1,1} = 10$, $p_{1,1} = 0.95$, $\tilde{w}_{1,1} = (4.32, 4.80, 5.28)$), and option 2, a larger and more reliable unit ($c_{1,2} = 20$, $p_{1,2} = 0.97$, $\tilde{w}_{1,2} = (4.95, 5.50, 6.05)$). Subsystems 2 and 3 each offer one option, the units above at cost 15. The target is $R_{\text{target}} = 0.90$. Design $[1, 1, 1]$ costs 40 and design $[2, 1, 1]$ costs 50. The trace is generated by `supp_trace_v11.py` (Section S2).

Phase 0: effective values and suffix tables. At $(0.5, 0)$ the threshold is $k^{0.5}(0) = k_U^{0.5} = 9.0$ and every effective capacity is a lower endpoint: $w_{1,1} = 4.56$, $w_{1,2} = 5.225$, $w_{2,1} = w_{3,1} = 4.75$. The largest capacity that subsystems 2 and 3 can add is $W_1^{\max} = 9.5$. Because these subsystems have one option each, $Z_1^{\text{suf}}(q) = 30$ for every $q \leq 9.5$.

Incumbent seed. The greedy design starts from the largest unit of every subsystem, $[2, 1, 1]$, whose accepted subsets are those with at least two units, so that $R_{\text{sys}}^{0.5,0}([2, 1, 1]) = 0.9784$. Replacing option 2 by the cheaper option 1 keeps the target met, $R_{\text{sys}}^{0.5,0}([1, 1, 1]) = 0.974$ (Step 4 above), and no further substitution lowers the cost. Phase 1 therefore starts with the incumbent $Z^* = 40$.

Phase 1: cost certification. Options are explored in ascending cost order, so the search first branches on option 1 of subsystem 1: $Z_{\text{next}} = 10$ and $q_{\text{next}} = 9.0 - 4.56 = 4.44$. The cost bound gives $Z_{\text{next}} + Z_1^{\text{suf}}(4.44) = 40$, which equals the incumbent and therefore does not prune: by Theorem 2(ii) only a strictly larger bound prunes, since an equal-cost completion may still win at the secondary level. The capacity bound $4.56 + 9.5 = 14.06 \geq 9.0$ and the optimistic reliability bound of Theorem 3, here $0.974 \geq 0.90$, do not prune either. The two single-option subsystems follow, and the leaf $[1, 1, 1]$ meets

the target, so $Z^* = \min(40, 40) = 40$. The search then returns to option 2 of subsystem 1: $Z_{\text{next}} = 20$, $q_{\text{next}} = 9.0 - 5.225 = 3.775$, and $20 + Z_1^{\text{suf}}(3.775) = 50 > 40$. Theorem 2(ii) prunes the subtree before its distribution is convolved, although its only design, $[2, 1, 1]$, is the more reliable one. Phase 1 ends with the certified minimum cost $Z^* = 40$.

Phase 2: iso-cost reliability maximization. With $Z^* = 40$ fixed, the traversal is repeated with the same tests. The only leaf of cost 40 is $[1, 1, 1]$, so $\mathbf{x}^* = [1, 1, 1]$ and $R^* = 0.974$, and option 2 of subsystem 1 is pruned again by the cost bound.

Result. The algorithm returns $(\mathbf{x}^*, Z^*, R^*) = ([1, 1, 1], 40, 0.974)$, the lexicographic optimum of Theorem 4: the cheapest design that meets the target, not the most reliable one. Without the seed, Phase 1 would start from $Z^* = \infty$, reach the same leaf first and prune option 2 in the same way.

S2 Computational Pipeline, Verification Checks and Environment

S2.1 Script-level provenance

1. `run_campaign_v2.py`: the 30-cell grid on each of the two derived instances (60 cells) solved with the interpreted backend; it is kept as a cross-check of the compiled backend, and the reported results of these instances come from `generate_benchmark_tables_v6.py`. Writes `campaign_v2_default.json`.

2. `run_measure_family.py`: closed forms for possibility, necessity and credibility, the chance-constraint equivalence sweep of Proposition 1, and the decomposition identity of Equation (13) of the main article.

3. `parametric_stability.py`: the complete parametric solution over $[0, 1]^2$, including the recovery check against the archived audit cells.

4. `run_scaling_campaign.py`: the unreduced interpreted scaling campaign (Regime 1 of Table S16) for $n \in \{8, 10, 12, 16, 20, 24, 28, 32\}$ with a 12 s budget per instance; it certifies every instance up to $n = 12$, and all larger instances time out. The cells $n = 14$ and $n = 18$ of that regime were measured separately under the same budget and are recorded, with a note, in `ablation_v6.json`.

5. `run_scaling_campaign_v4.py`: the synthetic scaling family under C++ and Theorem 1 reduction, with the reduced-versus-unreduced agreement verification.

6. `ablation_v5.py`: isolates the algorithmic dimensional reduction from compilation and budget increases at $n = 14$, computing iso-budget scaling frontiers.

7. `run_baseline_matched_v6.py`: the budget-matched genetic-algorithm comparison under both protocols.

8. `run_uncertainty_baseline_v6.py`: the oracle-substitution experiment against Liu's uncertainty theory, including the reproduction of the published worked example and the numerical admissibility check of the substituted bound.

9. `verify_pharma_oracle.py`: independent oracle verification on the 10-subsystem pharmaceutical plant benchmark.

10. `verify_fyffe_dp.py`: independent dynamic programming verification of the adapted Fyffe benchmark solutions.

11. `verify_fyffe_original.py`: reproduces the original published Fyffe et al. (1968) Table III optimum under their exact series-parallel model.

12. `verify_fyffe_provenance_v6.py`: the three provenance claims for the Fyffe benchmark: entry-by-entry agreement of the archived Table I with the published scan, reproduction of the Table III optimum under the source's own series-redundancy model, and the verbatim/synthetic breakdown of the adapted catalog. Writes `results/generated/fyffe_provenance_v6.json`.

13. `generate_benchmark_tables_v6.py`: the two benchmark result tables, rebuilt from the current solver with the necessity and possibility reliabilities of each prescribed design.

14. `run_audits_v6.py`: the verification checks (ii)–(ix) below: the closed-form grid comparison, the chance-constraint correspondence sweep, the exact-rational factorization check, the credibility audit under symmetric and asymmetric spreads, and the leaf/node accounting. Writes a single artifact, `results/generated/audits_v6.json`, recording the sampling domain, seed and outcome of each check.

15. `generate_figures_v6.py`: Figure S4 and the other figures of Sections S5 and S6, drawn from the artifacts, `benchmark_power_rts.json` and `benchmark_tables_v6.json`.

16. `plot_parametric_partition_v7.py`: the complete parametric solution of the IEEE RTS-79 fleet over the policy square; it checks that the closed-form boundaries of Corollary 2 lie on the level sets $\kappa = T$ and that every cell of the 30-cell grid falls in a region whose design and cost match the certified optimum of that cell.

17. `run_power_rts_campaign_v7.py`: the 30-cell exact optimization campaign on the IEEE RTS-79 power generation fleet case study. Writes `results/generated/benchmark_power_rts.json` and `results/generated/table_power_rts.tex`.

18. `run_power_rts_unit_specific_v7.py`: the unit-specific capacity-spread experiment on the IEEE RTS-79 fleet: all 30 policy cells solved by the unreduced exact search (time limit 600 s per cell), the 2×2 decomposition of each changed cell into thermal and hydro effects, and the hydro and thermal spread sweeps. Writes `results/generated/rts_unit_specific_v7.json` and the corresponding table. The 30 unit-specific solves take 1,363 s of solver time in total, the slowest 234 s; the decomposition and sweep runs are not timed.

19. `verify_unit_specific_bruteforce_v8.py`: an independent check of the unit-specific experiment: exhaustive enumeration of all 3^{10} fleets in exact rational arithmetic, sharing no code with the compiled search, for the common-spread optimum at $(0.3, 0.7)$, the unit-specific optima at $(0, 0)$, $(0.3, 0)$ and $(0.3, 0.7)$, and the \$171 M fleet selected under perturbed spreads, which it also confirms to be the crisp optimum of no integer threshold. Writes `results/generated/unit_specific_bruteforce_v8.json`.

20. `crisp_vs_fuzzy_v8.py`: the crisp and dual-fuzzy reliability of the seven RTS-79 designs of the parametric solution, with capacities only, demand only and both sides fuzzified, in the necessity cell and the optimistic boundary cell $(0, 1)$ (exact rational enumeration). Writes `results/generated/crisp_vs_fuzzy_v8.json` and `manuscript/table_crisp_vs_fuzzy.tex`.

21. `policy_menu_v8.py`: the design menu of the IEEE RTS-79 parametric solution (cost, necessity and possibility reliabilities and area share of each of the seven designs), the budget inversion and the γ intervals of Corollary 2 reported in Sections 5.1 and 6.1 of the main article. Writes `results/generated/policy_menu_v8.json` and `manuscript/table_policy_menu.tex`.

22. `run_delta_sensitivity_v7.py`: continuous sensitivity analysis over capacity spread $\delta \in [0.02, 0.20]$ and asymmetric seasonal thermal derating regimes on the IEEE RTS-79 fleet. Writes `results/generated/delta_sensitivity_v7.json` and generates `figures/fig_delta_sensitivity.pdf`.

23. `rts_true_frontier_v9.py` and `plot_true_frontier_v9.py`: exhaustive enumeration of all $3^{10} = 59{,}049$ candidate RTS-79 fleet designs (double precision) to construct the cost–necessity frontier (total cost Z against R_N) with and without the crisp adequacy constraint $R_{\text{modal}} \geq 0.95$, the crisp adequacy of the menu designs, the adequacy-constrained frontier of Table S11 (`manuscript/table_adequate_frontier.tex`) and Figure 2. Writes `results/generated/rts_true_frontier_v9.json` and generates `figures/fig_true_pareto_frontier.pdf`.

24. `replication_table_v10.py`: the replication table of Section S8 (case study and the two literature-derived instances), read from the archived grid and parametric artifacts; it checks that every grid optimum belongs to the parametric library. On the region of the policy square whose optimum is at no grid cell it reports the share on which the design of the nearest grid cell does not meet the reliability target, the largest cost excess of the cheapest grid design that does, and the area on which no grid design meets it. Each region $\{\lceil \kappa(\alpha, \gamma) \rceil = T\}$ is, at fixed $\alpha < 1$, the interval of γ between the boundaries of Corollary 2 for T and $T - 1$, so its area is a one-dimensional integral in α (midpoint rule, 2×10^5 nodes); whether a design meets the target at T is checked with its crisp reliability. Writes `results/generated/replication_v10.json` and `manuscript/table_replication.tex`.

25. `rts_rtarget_sensitivity_v9.py`: optimal fleet for $R_{\text{target}} \in \{0.90, 0.91, \ldots, 0.99\}$ at the effective thresholds $T = 148$, 180 and 220 MW, with the capacity left after the loss of the largest unit. Writes `results/generated/rts_rtarget_sensitivity_v9.json` and `manuscript/table_rts_rtarget_sensitivity.tex`.

26. `rts_unit_specific_matched_v9.py`: 30-cell exact rational brute-force optimization comparing common 10% spread, unit-specific RTS-79 spreads, and a matched common-spread control ($(\delta_l, \delta_r) = (0.06, 0.02)$ for all unit classes), compared with the unit-specific spreads. Writes `results/generated/rts_unit_specific_matched_v9.json`.

27. `baselines_v11.py`: the model baselines of Sections 4.2 and 5.3 of the main article and of Section S7. For the crisp model and the two one-sided models it reads the baseline design at its own effective threshold from the archived parametric library and evaluates it at the threshold of the dual-fuzzy model; the joint regions $\{\lceil \kappa_{\text{dual}} \rceil = t,\ \lceil \kappa_{\text{baseline}} \rceil = s\}$ are unions of γ-intervals with closed-form ends at fixed α and are integrated in α by the midpoint rule (2×10^4 nodes; the areas sum to one within 10^{-9}). For the probabilistic counterpart it computes the distribution $\Pr\{\hat{T} = T\}$ of the random effective threshold $\hat{T} = \lceil D/\Theta \rceil$ by adaptive quadrature, enumerates all 3^{10} designs of the RTS-79 and Pharmaceutical catalogs with exact capped convolutions, checks that the enumeration reproduces the parametric optimum at every integer threshold, and evaluates the selected design over the square. Writes `results/generated/baselines_v11.json`, `manuscript/table_baselines.tex` and `manuscript/table_baselines_supp.tex`.

28. `supp_trace_v11.py`: the optimization trace of Section S1, an independent transcription of Algorithm 1 of the main article in exact rational arithmetic for the worked example. Writes `results/generated/supp_trace_v11.json`.

29. `nesting_rts_example_v11.py`: Table S7 (Section S6.1); it maps the readings of Table 2 of the main article to their cells, computes the thresholds in exact rational arithmetic and reads the designs from the archived parametric solution of the fleet. Writes `results/generated/nesting_rts_example_v11.json`.

S2.2 Pipeline, audit checks and computational environment

The complete source code, including both solver backends and every experiment script, is hosted publicly at https://github.com/k-sarikaya/FuzzyWeighted. All computational artifacts are produced by a

deterministic pipeline.

Grid. The 30-cell grid evaluates $\alpha \in \{0.0, 0.3, 0.5, 0.7, 0.9, 1.0\}$ and $\gamma \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, yielding 30 exact optimization cells per benchmark. For the Pharmaceutical and Fyffe instances, results are archived in `benchmark_tables_v6.json` (60 cells); for the IEEE RTS-79 generation fleet, results are archived in `benchmark_power_rts.json` (30 cells), alongside the sensitivity and seasonal derating analysis in `delta_sensitivity_v7.json`. These cells also serve as the audit set against which the parametric solution of Section 5.1 of the main article is checked.

Experiment scripts. Every reported result is produced by a named script, and the mapping from each table and figure of the main article to the script that generated it, and the artifact that script writes, is given in the list above. The one exception is the pair of Regime 1 timeouts at $n = 14$ and $n = 18$, which were measured separately and are recorded in `ablation_v6.json`.

Reporting of search effort. Two distinct quantities describe how much of the search space an exact run touches, and they are reported separately throughout. "Leaves" counts complete designs evaluated and is comparable with 3^n; "nodes" counts partial assignments visited and is comparable with the full tree, $(3^{n+1} - 1)/2$. Dividing a node count by 3^n mixes the two and overstates the effort. Both backends seed the cost incumbent before Phase 1: the compiled backend with the greedy feasible design of Section 3.3 of the main article, the interpreted backend with a deterministic beam search (width 200, or 5,000 above ten subsystems). The reported node counts are those of the seeded search; a seed is a pruning bound only and cannot change the certified optimum.

Determinism and stochastic components. The exact solver is fully deterministic: identical inputs produce identical outputs across runs and platforms, and the seed field retained in the campaign file layout carries no stochastic meaning. The genetic-algorithm baseline is the only stochastic component in the paper; it is run with the five fixed seeds $0, \ldots, 4$, and per-seed outcomes rather than aggregates are reported in Section S9.

Exactness checks. Nine checks are applied. Checks (ii)–(ix) are produced by one auditing script, `run_audits_v6.py`, which writes one artifact recording the sampling domain, the seed and the outcome of each check, so that every check can be re-run; check (i) is recorded by `run_scaling_campaign_v4.py` in `scaling_campaign_v4.json`. (i) The reduced formulation must agree with the unreduced one on cost, reliability and design for every instance where the unreduced solve was evaluated within budget: across all 12 instances with $n \leq 14$, all 12 instances were checked, with all 12 agreeing identically on cost, reliability and design. (ii) The closed forms for possibility and necessity are compared against a brute-force sup–min construction on a refining grid; the error falls from 1.22×10^{-2} to 2.31×10^{-4}, an empirical order of 0.86, which is below one because the maximizer sits at a kink of the membership function and a grid can only approach it linearly. (iii) The chance-constraint correspondence of Proposition 1 is checked on 175,000 triples drawn with asymmetric spreads and asymmetric thresholds, in both the necessity and the possibility direction, with no discrepancy. (iv) The factorization underlying Theorem 1 is verified in exact rational arithmetic over an 8×7 grid of (α, γ) values evaluated across all three instances (IEEE RTS-79, Fyffe-derived and Pharmaceutical-derived): 4,592 comparisons across $3 \times (8 \times 7) = 168$ cells, with zero mismatch. (v) The coincidence of the central Hurwicz column with the credibility constraint $\mathrm{Cr} \geq 1/2$, discussed in Section 2.3 of the main article and in Section S10, is tested separately under symmetric and asymmetric spreads, the asymmetric sample being drawn to probe the case where the claim is expected to fail. Over 49,670 draws with symmetric spreads the central Hurwicz column and the credibility constraint never disagreed; over 50,330 draws with asymmetric capacity spreads they disagreed 1,390 times. One witness: at $\delta_l = 0.05$, $\delta_r = 0.4$, $\alpha = 0$ and modal capacity 40.61 against the symmetric threshold $\tilde{k} = (42.81, 46.40, 49.98)$, the central Hurwicz column accepts the subset while its credibility is only 0.354. (vi) The uncertain-measure oracle of Wei et al. (2025) is audited against the published worked example, reproducing the step-wise sums and $RM = 0.60$, and the root bound on the Pharmaceutical benchmark is recorded as 0.8488, confirming categorical infeasibility at the root node. (vii) The optimistic reliability upper bound of Theorem 3 is audited over 20,000 random prefix–completion pairs drawn across both benchmark catalogs, confirming zero violations with tightest observed slack bounded

by machine precision. (viii) The Arm-U uncertain-measure bound admissibility is audited over 20,000 random prefix–completion pairs, confirming zero violations (slack ≥ 0). (ix) Exhaustive evaluation of all $3^{10} = 59{,}049$ candidate designs under Arm-U confirms that the global maximum attainable uncertain measure is 0.848837, verifying that no design achieves the contractual target of 0.85. Corollary 1 of the main article, which maps every credibility chance constraint to a cell of the columns $\gamma = 0$ and $\gamma = 1$, is checked separately in `tests/test_membership.py`: 20,000 random comparisons with asymmetric capacity spreads and asymmetric thresholds, each at ten confidence levels on both sides of $1/2$, about 2×10^5 checks, with no discrepancy; replacing the cell $(2 - 2\lambda, 0)$ by $(1 - \lambda, 0)$ makes the check fail.

Computational environment. Two environments are used and are reported separately, because the runtimes are not comparable across them. The benchmark campaign and the analytic post-processing were run on a commodity desktop workstation (Microsoft Windows 11 64-bit, Intel Core i5-8265U, 8 GB RAM) under Python 3.12.5 with NumPy 2.1.1 and Matplotlib 3.9.2. The scaling family, the ablation and the genetic-algorithm comparison were run on a Linux host (kernel 6.18, Intel Xeon 2.10 GHz, 8 GB RAM) under Python 3.11.15, with the exact recursion compiled by `g++ 13.3.0` at `-O3 -std=c++17`. The interpreted and compiled backends are checked against one another on every instance both can solve. The runtimes and node counts reported for the exact solver are those of the compiled backend, except Regime 1 of Table S16; the uncertain-measure arm of Section S4.2 and the genetic algorithm are interpreted. Both backends compute capacities in exact rational arithmetic, scaled to integers, and compare reliabilities with the tolerance $\epsilon = 10^{-12}$. Because the genetic algorithm is interpreted while the exact solver is compiled, wall-clock matching in Protocol A favors the exact solver; this is the reason Protocol B reports an evaluation budget alongside it. No GPU acceleration is used and the solver is CPU-bound.

S3 Verification of the Uncertainty-Theory Oracle

The comparison in Section S4.2 substitutes the uncertain-measure evaluator of Wei, Shi and Zhang (2025, *Symmetry* 17:908) for the possibilistic one. Because the comparison rests on that substitution, the implementation was verified against the worked example published with the original method before being used, and the record is given here.

S3.1 Reproduction of the published worked example

The published example has ten components with a crisp threshold $k = 24$, state measures $v = (0.5, 0.6, 0.7, 0.5, 0.9, 0.9, 0.8, 0.9, 0.7, 0.8)$ and linear uncertain weights $\mathcal{L}(1, 2.2), \ldots, \mathcal{L}(3, 4.6)$. The source reports the reliability measure as $RM = 0.6$ by its traverse scheme at step 0.1 and as 0.5977 by bisection; our implementation returns $RM = 0.6$ by both schemes.

Table S1: Partial sums $\sum_i \Phi_i^{-1}(1-\nu) \nVdash_i(\nu)$ of the published example.

ν	0.9	0.8	0.7	0.6
Published	8.30	15.40	21.64	24.72
Computed here	8.30	15.12	21.22	24.16

Table S1 compares the partial sums. The value at $\nu = 0.9$ agrees exactly. The remaining three differ, and the discrepancy is not an artifact of our reading of the method: exhaustive enumeration of all 2^{10} subsets of the published component table shows that *no* subset of those weights produces the published figures at $\nu = 0.7$ or $\nu = 0.6$ under any evaluation level. We therefore treat those three intermediate values as arithmetic slips in the source. The quantity that the method reports, $RM = 0.6$, is reproduced, and the orientation of the step function used here (component i is counted when $\nu \leq v_i$) is the one stated in the multi-state companion paper (*Symmetry* 17:912) and the only one consistent with the published result.

S3.2 Admissibility of the substituted bound

The exact search under the uncertain measure requires an admissible optimistic bound. Assigning to each unassigned subsystem the largest state measure and the largest capacity in its catalog yields a system that attains at least as much weight at every confidence level; monotonicity of the uncertain measure in each argument then makes the resulting value an upper bound on every completion. This was checked numerically on 20,000 prefix–completion pairs drawn at random from the Pharmaceutical catalog: no violation occurred, and the tightest observed slack was exactly zero, attained when the prefix is complete.

S4 Pharmaceutical-Derived Instance and Uncertainty-Theory Comparison

S4.1 The Pharmaceutical-derived instance

The Pharmaceutical Plant study of Garg and Sharma (2013) models a 10-subsystem industrial manufacturing network (spanning unit operations from weighing, sifting, and granulation to tablet compression, coating, and strip packing). In the source paper, the problem is formulated over continuous component reliabilities with exponential cost curves and continuous physical weight constraints. To instantiate a discrete redundancy allocation problem, an adapted synthetic catalog was constructed whose topology is directly anchored in the physical plant: the capacities of the first option of each subsystem, $w_i = (9, 7, 5, 9, 9, 10, 6, 5, 8, 6)$, are numerically set to the component physical weights in Table 1 of Garg and Sharma (2013); the second option is a synthetic higher-capacity variant, one or two units larger, and all costs and reliabilities are synthetic choices designed to induce non-trivial combinatorial redundancy trade-offs (the source problem had continuous reliabilities). The benchmark parameters are summarized in Table S2.

Table S2: Parameters of the Pharmaceutical-derived instance. The nominal capacities w_i of the first option are adapted from Table 1 of Garg and Sharma (2013); all other values are synthetic.

Parameter	Pharmaceutical (adapted)
Subsystems n	10
Options per subsystem (incl. null)	3
Design space	$3^{10} = 59{,}049$
Fuzzy threshold $\tilde{k}$	(34, 40, 46)
Reliability target R_{target}	0.85
Capacity spread (δ_l, δ_r)	(0.1, 0.1)
Options verbatim from source	None (synthetic discrete catalog)

Table S3 documents the minimum-cost certified designs across all 30 policy cells of the Pharmaceutical-derived instance. The certified minimum cost scales from 92 in the optimistic boundary cell $(0.0, 1.0)$ to 158 in the conservative cell $(0.3, 0.0)$.

Infeasibility of Cell $(0, 0)$. In contrast to the Fyffe-derived instance and the IEEE RTS-79 fleet, where all 30 cells admit feasible configurations, cell $(0.0, 0.0)$ on the Pharmaceutical catalog is certified to be infeasible (marked with †). Under the extreme conservative stance ($\alpha = 0.0, \gamma = 0.0$), the required threshold becomes $T = \lceil 46/(1 - 0.10) \rceil = \lceil 46/0.90 \rceil = 52$. If all 10 subsystems are installed with their maximum-capacity options, the total installed modal capacity is $86 \geq 52$. Thus, capacity sufficiency is attainable; the binding obstacle is stochastic reliability. An exhaustive combinatorial evaluation of all $3^{10} = 59{,}049$ candidate configurations proves that the maximum achievable system reliability at threshold $T = 52$ is 0.8156, which fails to meet the prescribed adequacy requirement $R_{\text{target}} = 0.85$. (This finding is independently audited in `results/generated/audits_v6.json` under `pharma_00_infeasibility`).

Table S3: Exact global optimum over the (α, γ) grid, adapted Pharmaceutical Plant catalog. T is the effective integer threshold of Theorem 1 of the main article; R_{sys} is the reliability of the prescribed design under the Hurwicz rule at that cell, and R_N, R_Π are the necessity and possibility reliabilities of the same design. † marks a cell in which the exact solver certifies that no design meets the target.

α	γ	T	Cost	R_{sys}	R_N	R_Π	Nodes	Time (s)
0.0	0.0	52			† no feasible design			
0.3	0.0	48	158	0.8618	0.882	0.979	1,904	0.00
0.5	0.0	46	151	0.8506	0.829	0.969	5,762	0.00
0.7	0.0	44	144	0.8507	0.750	0.951	9,188	0.00
0.9	0.0	42	134	0.8519	0.711	0.934	12,797	0.01
1.0	0.0	40	123	0.8625	0.599	0.893	15,026	0.01
0.0	0.3	45	146	0.8523	0.775	0.958	7,124	0.00
0.3	0.3	43	139	0.8563	0.769	0.951	11,069	0.00
0.5	0.3	43	139	0.8563	0.769	0.951	11,069	0.00
0.7	0.3	42	134	0.8519	0.711	0.934	12,797	0.00
0.9	0.3	41	132	0.8521	0.680	0.924	16,094	0.00
1.0	0.3	40	123	0.8625	0.599	0.893	15,026	0.00
0.0	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
0.3	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
0.5	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
0.7	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
0.9	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
1.0	0.5	40	123	0.8625	0.599	0.893	15,026	0.00
0.0	0.7	37	115	0.8544	0.472	0.841	23,711	0.01
0.3	0.7	38	117	0.8546	0.511	0.864	20,414	0.00
0.5	0.7	39	120	0.8581	0.552	0.879	17,780	0.00
0.7	0.7	39	120	0.8581	0.552	0.879	17,780	0.00
0.9	0.7	40	123	0.8625	0.599	0.893	15,026	0.00
1.0	0.7	40	123	0.8625	0.599	0.893	15,026	0.00
0.0	1.0	31	92	0.8578	0.151	0.594	23,243	0.01
0.3	1.0	34	103	0.8836	0.345	0.737	23,765	0.01
0.5	1.0	36	111	0.8533	0.400	0.795	24,791	0.01
0.7	1.0	38	117	0.8546	0.511	0.864	20,414	0.01
0.9	1.0	40	123	0.8625	0.599	0.893	15,026	0.00
1.0	1.0	40	123	0.8625	0.599	0.893	15,026	0.00

S4.2 Oracle Substitution: Possibility Theory vs. Liu's Uncertainty Theory

The comparison between Possibility Theory and Liu's Uncertainty Theory is evaluated empirically by holding the decision problem fixed and changing only the object that grades a design.

Setup. The catalogs, the cost objective, the confidence requirement and the lexicographic rule are identical in both arms. Arm P evaluates a design with the reliability of the Hurwicz rule (Section 2 of the main article); Arm U evaluates it with the uncertain measure of Wei et al. (2025), whose Theorem 4 reduces the reliability of a weighted k-out-of-n system to the root of a one-dimensional monotone condition. Each option's reliability is read as the uncertain measure of its component state, and each capacity TFN $(w(1-\delta_l),\, w,\, w(1+\delta_r))$ is carried over as the zigzag uncertain variable on the same three parameters, the natural image of a triangular fuzzy number in that calculus. Both arms are solved to certified optimality by a lexicographic branch and bound, so a difference in outcome is attributable to the calculus rather than to the search. For Arm U the admissible bound is established by substituting, for each unassigned subsystem, the largest state measure and the largest capacity in its catalog.

A caution about the mapping. Translating p_i into an uncertain state measure and a TFN into a zigzag variable is a change of formalism, not an identity. An uncertain measure and a possibility distribution are different objects on different scales, and requiring both to clear the same nominal level is a convention adopted for comparability, not a claim that the two requirements are equivalent.

Table S4: Oracle substitution on the two benchmark catalogs at the modal threshold. "Leaves" counts complete designs evaluated; "nodes" counts search-tree expansions. Runtimes are not comparable across arms (Arm P is compiled, Arm U interpreted) and are reported only for completeness.

Catalog	Arm	Cost	Reliability	Nodes	Leaves	Time (s)
Pharmaceutical (3^{10})	P	123	0.8625	1.50×10^4	9	0.004
	U	infeasible		1	0	< 0.01
Fyffe (4^{14})	P	23	0.9224	1.03×10^7	760	2.08
	U	22	0.9200	2.39×10^6	6.84×10^4	72.4

Findings. The two arms disagree on both catalogs (Table S4), and they disagree in different ways:

- On the Pharmaceutical catalog the disagreement is categorical. Arm P returns a feasible design at cost 123 with reliability 0.8625 against a target of 0.85. Arm U declares the instance infeasible, and the declaration is not an artifact of pruning: the largest uncertain measure attainable by *any* of the 59,049 designs is 0.8488, which the root bound reproduces exactly. The cause is structural. At a confidence level ν the uncertain measure credits only the units whose state measure is at least ν. At $\nu = 0.85$ these are units of five subsystems (2, 3, 4, 6 and 10), and their capacity credited at that level is 39.99 against the threshold 40; the best design installs exactly these units and certifies 0.8488. A unit with a lower state measure adds nothing at the required level, however much capacity it carries. The possibilistic reliability, by contrast, aggregates subset probabilities, so every installed unit contributes to the probability of the accepted subsets. The two frameworks thus disagree on whether the design problem has a feasible solution.

- On the Fyffe catalog both arms find designs, but not the same one: Arm P selects a design costing 23 and Arm U one costing 22, differing in three of the fourteen subsystems. Adopting the uncertainty calculus would therefore lead to a different system at a lower cost; the two numbers answer different questions.

Computational Cost. Arm U expands fewer search nodes than Arm P on the Fyffe catalog (2.39×10^6 against 1.03×10^7). The difference lies in leaf evaluations: Arm P evaluates 760 complete designs where Arm U evaluates 68,384, so in leaf evaluations the possibilistic bound is nearly two orders of magnitude tighter, and each possibilistic evaluation is a capped convolution rather than a bisection.

Summary. Even at nominal demand, the choice of calculus changes which design is prescribed, what it costs and, on the Pharmaceutical catalog, whether a feasible design exists. The calculus used to grade designs should therefore be stated explicitly.

S5 The Fyffe-Derived Instance

The main article uses the Fyffe-derived instance to check whether the findings of the case study recur (Section 5.1 of the main article). This section gives the adaptation, the results on the 30-cell grid, and further results that repeat on a larger catalog the patterns reported for the IEEE RTS-79 fleet.

S5.1 Adaptation of the source problem

Fyffe et al. (1968) pose a series system of fourteen functional units. For each unit one selects a design alternative (R, C, W) and an integer number m of identical units placed in parallel, so that the unit achieves $1 - (1 - R)^m$; the objective is to maximize the product of unit reliabilities subject to a total cost budget $C \leq 130$ and a total *physical weight* budget $W \leq 170$. The model of the main article is a weighted k-out-of-n:G design problem, in which each subsystem contributes an operational capacity to an aggregate that must clear a threshold and the objective is minimum cost subject to a reliability target. Four changes were required.

1. **The third field is reinterpreted.** In the source, W is a physical weight consumed against a budget; here it is the operational capacity w_i that contributes to the success threshold. This is a reinterpretation of the data, not a translation of the problem.

2. **Parallel redundancy is removed.** The integer multiplicity m of the source is dropped; each subsystem selects exactly one option from a finite catalog.

3. **A null option is added.** Every subsystem gains a $(0, 0, 0)$ "not installed" option, which the source does not admit (there $m \geq 1$ always).

4. **The source budgets are not enforced.** $C \leq 130$ and $W \leq 170$ play no role; cost is minimized rather than bounded, and capacity is required rather than limited.

Each of the fourteen subsystems carries three non-null options. Thirty-one of these 42 options are verbatim rows of Table I of Fyffe et al. (1968): for every unit the first two options are its published alternatives 1 and 2, and for units 5, 7 and 12 the third option is the published alternative 3 as well. The remaining eleven third options are synthetic, constructed to give each subsystem a higher-reliability, higher-cost choice. The published alternative 4, where it exists, is not used.

Table I of Fyffe et al. (1968) was transcribed from the published scan into a machine-readable archive and verified entry by entry against it; all fourteen units match. The *original* problem (series architecture, parallel redundancy, both budgets) was then solved with an exact dynamic program over integer cost–weight states. It returns $C = 119$, $W = 170$ and $R = 0.97001500$, matching the published optimum to the precision at which it is reported (Table S5), and the recovered policy agrees with Table III of the source in all fourteen units, both in the design alternative selected and in the number of parallel units.

Table S5: Independent reproduction of the original Fyffe et al. (1968) allocation problem from the archived Table I data, under the source's own series-redundancy model.

	Total cost	Total weight	System reliability
Fyffe et al. (1968), Table III	119	170	0.9700
Exact reproduction here	119	170	0.9700

S5.2 Further results

Table S6: Exact global optimum over the (α, γ) grid, adapted Fyffe 14-subsystem catalog. T is the effective integer threshold of Theorem 1 of the main article; R_{sys} is the reliability of the prescribed design under the Hurwicz rule at that cell, and R_N, R_Π are the necessity and possibility reliabilities of the same design (all 30 cells admit feasible certified optima).

α	γ	T	Cost	R_{sys}	R_N	R_Π	Nodes	Time (s)
0.0	0.0	67	36	0.9042	0.976	1.000	24,281,814	6.52
0.3	0.0	62	32	0.9097	0.931	0.999	28,277,062	6.66
0.5	0.0	58	28	0.9001	0.813	0.996	17,436,386	3.97
0.7	0.0	55	26	0.9119	0.736	0.993	14,234,490	3.10
0.9	0.0	52	24	0.9075	0.569	0.983	11,746,022	2.46
1.0	0.0	50	23	0.9224	0.485	0.974	10,313,586	2.13
0.0	0.3	57	28	0.9219	0.816	0.997	21,976,334	4.83
0.3	0.3	55	26	0.9119	0.736	0.993	14,234,490	3.27
0.5	0.3	54	26	0.9224	0.693	0.992	17,489,986	3.79
0.7	0.3	52	24	0.9075	0.569	0.983	11,746,022	2.36
0.9	0.3	51	24	0.9294	0.549	0.981	13,723,426	2.80
1.0	0.3	50	23	0.9224	0.485	0.974	10,313,586	2.10
0.0	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.03
0.3	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.01
0.5	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.14
0.7	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.00
0.9	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.10
1.0	0.5	50	23	0.9224	0.485	0.974	10,313,586	2.08
0.0	0.7	45	20	0.9302	0.233	0.923	6,608,758	1.16
0.3	0.7	46	20	0.9162	0.233	0.923	6,497,834	1.16
0.5	0.7	48	21	0.9036	0.315	0.951	6,675,182	1.26
0.7	0.7	49	22	0.9133	0.376	0.963	8,036,886	1.53
0.9	0.7	50	23	0.9224	0.485	0.974	10,313,586	2.09
1.0	0.7	50	23	0.9224	0.485	0.974	10,313,586	2.14
0.0	1.0	37	14	0.9031	0.000	0.428	1,056,214	0.16
0.3	1.0	41	17	0.9295	0.040	0.808	2,724,778	0.47
0.5	1.0	43	18	0.9061	0.110	0.836	3,357,026	0.72
0.7	1.0	46	20	0.9162	0.233	0.923	6,497,834	1.26
0.9	1.0	49	22	0.9133	0.376	0.963	8,036,886	1.60
1.0	1.0	50	23	0.9224	0.485	0.974	10,313,586	2.07

Table S6 reports the certified optimum of every one of the 30 policy cells; all of them admit a feasible design. The minimum cost ranges from 14 in the optimistic boundary cell $(0.0, 1.0)$ to 36 in the necessity cell $(0.0, 0.0)$.

Cost and the strength of the guarantee. Figure S1 is the Fyffe counterpart of Figure S3 and of Figure S4. Under $\gamma = 0$ the cost escalates from 23 at $\alpha = 1$ to 36 at $\alpha = 0$; under $\gamma = 1$ the same widening lowers it to 14. All lines meet at $\alpha = 1$ and the line $\gamma = 0.5$ is flat. Along the optimistic column the crisp baseline at $(1, 1)$ costs 23 and delivers necessity 0.485, whereas the cell $(0, 1)$ costs 14 and delivers necessity 0.000; the intermediate cells interpolate monotonically: cost 17 at necessity 0.04, 18 at 0.11, 20 at 0.23, 22 at 0.38; the nine units of capital saved are bought at roughly 0.054 of necessity per unit.

Cost and configuration over the grid. Figure S2 maps the certified minimum cost and the number of installed subsystems over the 30 cells. The installed count ranges from 13 at $(0, 0)$ to 7 at $(0, 1)$: the declared attitude changes both the cost of the design and the redundancy it carries.

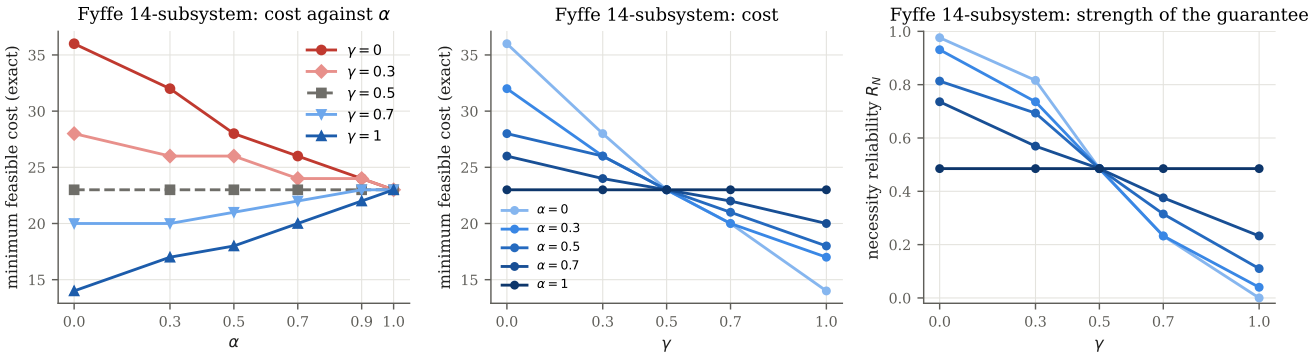


Figure S1: Fyffe-derived instance. Left: minimum feasible cost against α, one line per γ. Center and right: minimum feasible cost and necessity reliability R_N of the prescribed design against γ, one line per α ($\alpha = 0.9$ omitted for legibility). Every point is the certified optimum of its cell.

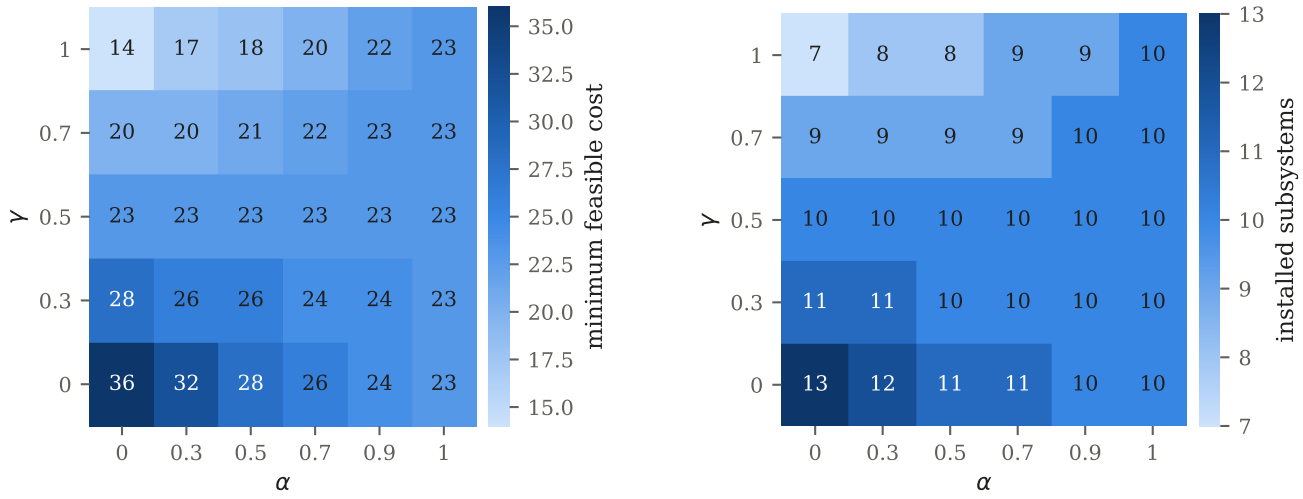


Figure S2: Fyffe-derived instance. Left: certified minimum cost $Z^*(\alpha, \gamma)$. Right: number of subsystems installed by the certified optimum of each cell.

S6 Further Results on the IEEE RTS-79 Case Study

S6.1 The classical readings on the fleet

Table 2 of the main article places the classical readings of the success condition at cells of the policy square. Table S7 applies it to the fleet: each reading gives a cell, the cell gives the effective threshold T, and the threshold gives the optimal design of the complete parametric solution, so no optimization is solved. The crisp model buys D5 for \$180 M. Requiring necessity 0.7 or 0.9 moves the threshold to 208 or 216 MW and buys D7 for \$203 M, the same fleet as the interval worst case; credibility 0.8 stops at D6, and the possibility and low-credibility requirements are met by D4 for \$158 M.

Table S7: The readings of Table 2 of the main article on the IEEE RTS-79 fleet ($\tilde{k} = (162, 180, 198)$ MW, capacity spread $\delta = 0.10$). Each reading is mapped to its cell; κ and $T = \lceil \kappa \rceil$ are the effective continuous and integer thresholds in MW, and the design is read from the complete parametric solution (designs D1–D7 of Table 4 of the main article). No optimization is solved for this table.

Reading	Condition	Cell	κ	T	Design	Cost (M\$)	R_N
Crisp model (modal values)	$W_S \geq k_2$	$\alpha = 1$	180.00	180	D5	180	0.506
Interval worst case	$W^0_{S,L} \geq k^0_U$	(0, 0)	220.00	220	D7	203	0.951
Interval best case	$W^0_{S,U} \geq k^0_L$	(0, 1)	147.27	148	D1	130	0.000
Necessity	$N(\tilde{W}_S \geq \tilde{k}) \geq 0.7$	(0.3, 0)	207.10	208	D7	203	0.951
Necessity	$N(\tilde{W}_S \geq \tilde{k}) \geq 0.9$	(0.1, 0)	215.60	216	D7	203	0.951
Possibility	$\Pi(\tilde{W}_S \geq \tilde{k}) \geq 0.7$	(0.7, 1)	169.51	170	D4	158	0.000
Credibility	$\mathrm{Cr}(\tilde{W}_S \geq \tilde{k}) \geq 0.3$	(0.6, 1)	166.15	167	D4	158	0.000
Credibility	$\mathrm{Cr}(\tilde{W}_S \geq \tilde{k}) \geq 0.8$	(0.4, 0)	202.98	203	D6	180	0.617

S6.2 Minimum cost against the cut level

Figure S3 plots the certified minimum cost of the RTS-79 fleet over the 30-cell grid against α, one line per γ; the values are those of Table 5 of the main article, and the pattern is discussed in its Section 5.2.

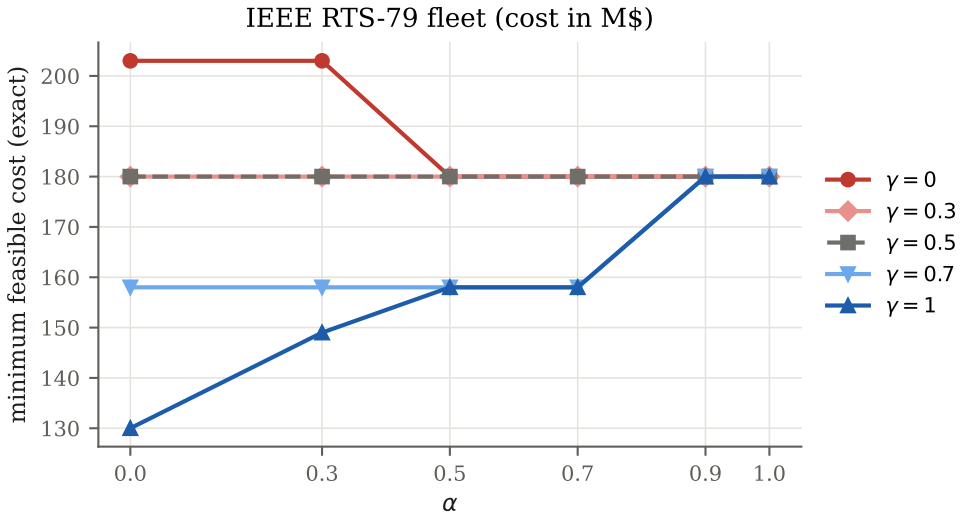


Figure S3: Minimum feasible cost against α, one line per γ, for the IEEE RTS-79 fleet. Every point is the certified optimum of its cell; lines overlap where costs coincide. The Fyffe counterpart is Figure S1.

S6.3 Catalog of the fleet

Table S8 lists the complete discrete catalog of the case study of Section 4.1 of the main article. Option 0 is the null option; hydro units are available only at Bays 3–6.

Table S8: Complete 10-bay discrete option catalog for the IEEE RTS-79 generation fleet case study. Option 0 represents an uncommitted (null) bay ($w = 0, c = 0, p = 0$). Unit classes and availability parameters $p_{i,j} = 1 - \mathrm{FOR}_i$ are from RTS-79; capital costs are illustrative engineering estimates.

Bay i	**Option** j	**Generating Unit Class**	**Capacity** $w_{i,j}$ (MW)	**Capital Cost** $c_{i,j}$ (\$M)	**Availability** $p_{i,j}$
1–2	1	12 MW Oil Steam	12	14	0.980
	2	20 MW Combustion Turbine	20	18	0.900
3–4	1	20 MW Combustion Turbine	20	18	0.900
	2	50 MW Hydro	50	45	0.990
5–6	1	50 MW Hydro	50	45	0.990
	2	76 MW Coal Steam	76	68	0.980
7–8	1	76 MW Coal Steam	76	68	0.980
	2	100 MW Oil Steam	100	85	0.960
9–10	1	100 MW Oil Steam	100	85	0.960
	2	155 MW Coal Steam	155	135	0.960

S6.4 Reliability target

At $R_{\text{target}} = 0.95$ the cost-minimizing fleets of the optimistic boundary, modal and necessity cells are series arrangements at their threshold T: the loss of any single unit drops the working capacity below T. Table S9 repeats the optimization for $R_{\text{target}} \in \{0.90, 0.91, \ldots, 0.99\}$ at the thresholds of the optimistic boundary cell $(0, 1)$ ($T = 148$ MW), the modal cell $(0.5, 0.5)$ ($T = 180$ MW) and the necessity cell $(0, 0)$ ($T = 220$ MW).

Table S9: Sensitivity of the optimal fleet to the reliability target $R_{\text{target}} \in \{0.90, 0.91, \ldots, 0.99\}$ on the IEEE RTS-79 fleet at three effective thresholds. Consecutive targets with the same optimal design are merged. R_{sys} is the reliability at the effective threshold T. The last column states whether the working capacity still reaches T after the loss of the largest installed unit, and gives that remaining capacity.

Regime	R_{target}	Optimal fleet	MW	Cost (M\$)	R_{sys}	Largest-unit loss
Optimistic boundary $(0, 1)$, $T = 148$ MW	0.90–0.95	1 × 50 MW Hydro + 1 × 100 MW Oil	150	130	0.9504	No (50 MW)
	0.96–0.97	3 × 50 MW Hydro	150	135	0.9703	No (100 MW)
	0.98–0.99	4 × 50 MW Hydro	200	180	0.9994	Yes (150 MW)
Modal $(0.5, 0.5)$, $T = 180$ MW	0.90–0.92	1 × 76 MW Coal + 1 × 100 MW Oil + 1 × 12 MW Oil	188	167	0.9220	No (88 MW)
	0.93–0.94	2 × 50 MW Hydro + 1 × 76 MW Coal + 1 × 12 MW Oil	188	172	0.9413	No (112 MW)
	0.95–0.96	4 × 50 MW Hydro	200	180	0.9606	No (150 MW)
	0.97–0.99	4 × 50 MW Hydro + 1 × 12 MW Oil + 1 × 20 MW CT	232	212	0.9948	Yes (182 MW)
Necessity $(0, 0)$, $T = 220$ MW	0.90–0.93	1 × 50 MW Hydro + 1 × 76 MW Coal + 1 × 100 MW Oil	226	198	0.9314	No (126 MW)
	0.94–0.95	3 × 50 MW Hydro + 1 × 76 MW Coal	226	203	0.9509	No (150 MW)
	0.96–0.97	4 × 50 MW Hydro + 1 × 76 MW Coal	276	248	0.9794	No (200 MW)
	0.98–0.99	4 × 50 MW Hydro + 1 × 76 MW Coal + 1 × 20 MW CT	296	266	0.9967	Yes (220 MW)

A fleet that still reaches T after the loss of its largest unit becomes optimal at $R_{\text{target}} = 0.97$ in the modal cell (\$212 M, 232 MW, $R_{\text{sys}} = 0.9948$) and at $R_{\text{target}} = 0.98$ in the other two cells (\$180 M and 200 MW in the optimistic boundary cell; \$266 M and 296 MW in the necessity cell). For lower targets no optimal fleet tolerates the loss of its largest unit. In the modal cell three designs are optimal below that

point: \$167 M up to $R_{\text{target}} = 0.92$, \$172 M at 0.93–0.94 and \$180 M at 0.95–0.96. These breakpoints are properties of this catalog and target range, not general rules.

S6.5 Unit-specific capacity spreads

This section gives the full results summarized in Section 5.5 of the main article.

The baseline RTS-79 formulation of Section 4.1 of the main article gives every candidate unit the same proportional spread $\delta = 0.10$, which is the hypothesis under which Theorem 1 holds. The sources of capacity imprecision do not, however, affect all generating technologies alike, and for one of them RTS-79 is explicit. Its Table 6 gives the available capacity of the hydro units as 100% of the 50 MW rating in the first two quarters of the year and 90% in the last two, so the hydro capacity is the triangular number $(45, 50, 50)$ MW, that is $(\delta_l, \delta_r) = (0.10, 0)$: the same downside as the common spread and no upside. For thermal units RTS-79 reports full outages only and leaves partial outages to the user (note to its Table 4), so we assume spreads that encode the qualitative effect of ambient temperature: $(0.15, 0.05)$ for the 20 MW combustion turbines, whose output falls as combustion air density drops, and $(0.05, 0.02)$ for the oil- and coal-fired steam units (12, 76, 100 and 155 MW), which are limited by condenser back-pressure. The thermal values are an assumption of this study, chosen to reflect the qualitative ordering rather than measured derating curves; how much the results depend on them is reported below.

With heterogeneous spreads the hypothesis of Theorem 1 fails. The effective threshold is no longer a single closed-form function of (α, γ), so each policy cell must be solved by the unreduced exact search on the component-specific α-cut intervals $[w^{\alpha}_{i,L}, w^{\alpha}_{i,U}]$. With a time limit of 600 s per cell, all 30 cells complete; the slowest takes 234 s, whereas every cell of the reduced formulation solves in at most 0.01 s of solver time (Section 5.2 of the main article). As a check on the implementation, the unreduced search with every unit spread set to 0.10 reproduces the reduced optimum in each of the three cells tested, and an independent exhaustive enumeration of all 3^{10} fleets in exact rational arithmetic reproduces four of the optima reported below and the \$171 M fleet discussed at the end of this section (Section S2). Table S10 lists every cell in which the optimum changes. It also decomposes each change: in the "thermal only" column the thermal units receive their unit-specific spreads while hydro keeps the common spread, and in the "hydro only" column the reverse.

Table S10: Policy cells of the IEEE RTS-79 fleet in which unit-specific capacity spreads change the optimal fleet relative to the common spread $\delta = 0.10$. Unit-specific spreads: hydro $(\delta_l, \delta_r) = (0.10, 0)$ from RTS-79 Table 6; combustion turbine $(0.15, 0.05)$ and steam units $(0.05, 0.02)$, an assumption of this study. "Thermal only" changes the thermal units and keeps hydro at the common spread; "hydro only" does the reverse.

α	γ	Common-spread fleet	Unit-specific fleet	Common	Thermal only	Hydro only	Unit-specific
0.3	0.0	3 × 50 MW hydro + 1 × 76 MW coal	1 × 50 MW hydro + 1 × 155 MW coal	\$203 M	\$180 M	\$203 M	\$180 M
0.0	0.3	4 × 50 MW hydro	1 × 50 MW hydro + 1 × 155 MW coal	\$180 M	\$180 M	\$180 M	\$180 M
0.3	0.7	2 × 50 MW hydro + 1 × 76 MW coal	4 × 50 MW hydro	\$158 M	\$158 M	\$158 M	\$180 M
0.5	0.7	2 × 50 MW hydro + 1 × 76 MW coal	4 × 50 MW hydro	\$158 M	\$158 M	\$171 M	\$180 M
0.7	0.7	2 × 50 MW hydro + 1 × 76 MW coal	4 × 50 MW hydro	\$158 M	\$180 M	\$180 M	\$180 M
0.0	1.0	1 × 50 MW hydro + 1 × 100 MW oil	3 × 50 MW hydro + 1 × 12 MW oil	\$130 M	\$135 M	\$135 M	\$149 M
0.3	1.0	3 × 50 MW hydro + 1 × 12 MW oil	2 × 50 MW hydro + 1 × 76 MW coal	\$149 M	\$149 M	\$158 M	\$158 M

Note: 7 of 30 cells differ; 0 timed out at 600 s. In 7 of the 7 differing cells the unit-specific optimum coincides with the crisp optimum of some integer threshold (library: 73 thresholds, 7 distinct designs). Unreduced solve time per cell: 0.008–234 s; reduced formulation: ≤ 0.01 s per cell.

The optimal fleet changes in 7 of the 30 cells and its cost in 6, in both directions.

- *Necessity side: over-investment.* At $(0.3, 0.0)$ the common spread prescribes three hydro units and one 76 MW coal unit at \$203 M, and the unit-specific model one hydro unit and one 155 MW coal unit at \$180 M, a saving of \$23 M (11.3%). The saving is due to the thermal spreads alone; the hydro values of Table 6 change nothing there. In the necessity cell $(0.0, 0.0)$ the optimum is unchanged, because on the downside a hydro unit is worth 45 MW under Table 6 exactly as under the common spread. At $(0.0, 0.3)$ the fleet changes at equal cost.

- *Possibility side: under-investment.* In five cells the common spread lets upper-bound capacities satisfy the requirement too cheaply, and the unit-specific model requires more capital: +\$19 M at $(0.0, 1.0)$, +\$9 M at $(0.3, 1.0)$, and +\$22 M at $(0.3, 0.7)$, $(0.5, 0.7)$ and $(0.7, 0.7)$. The main driver is the missing hydro upside: the common spread credits a hydro unit with up to 55 MW, Table 6 with at most 50. The decomposition shows that the two sources interact. At $(0.3, 1.0)$ the hydro change alone accounts for the difference and at $(0.7, 0.7)$ either change alone suffices, but at $(0.0, 1.0)$ each change on its own raises the cost only to \$135 M, and at $(0.3, 0.7)$ neither change on its own moves the optimum; the full increase arises only when both are applied.

In the remaining 23 cells the prescribed fleet is unchanged.

Matched common-spread controls. To check whether the seven changed optima arise from unit-level heterogeneity or from a smaller aggregate spread, we evaluated all 30 cells against a single common-spread control $(\delta_l, \delta_r) = (0.06, 0.02)$ applied to every unit. Under the matched control, exactly 5 of the 7 discrepancies with the baseline common spread are reproduced identically. The two remaining cells are $(0.0, 0.3)$ and $(0.3, 0.7)$. In $(0.0, 0.3)$ the cost is unchanged (\$180 M) but the fleet is not: the unit-specific hydro spread has no upside, so at the effective demand $k^0(0.3) = 187.2$ MW four hydro units supply only $4 \times (0.3 \cdot 50 + 0.7 \cdot 45) = 186$ MW, and the equal-cost D6 replaces D5. Cell $(0.3, 0.7)$ is the least robust cell of the table (see below). With this one control, most of the changes are reproduced by a smaller common spread; separating magnitude from heterogeneity in general would require a sweep over common spreads.

Sensitivity to the spread values. Varying the hydro downside ($\delta_l \in \{0, 0.02, 0.05, 0.10\}$, with $\delta_r = 0$) leaves the optimum unchanged in six of the seven cells; at $(0.3, 0.7)$ the cost is \$158 M for $\delta_l \leq 0.02$ and \$180 M from 0.05. Varying the assumed thermal spreads (the combustion-turbine downside over $\{0.10, 0.15, 0.20\}$ or the steam downside over $\{0.02, 0.05, 0.08\}$) leaves the optimum unchanged in six cells, including the necessity-side saving at $(0.3, 0.0)$. The exception is again $(0.3, 0.7)$: with a combustion-turbine downside of 0.10, or a steam downside of 0.02, the optimum becomes a \$171 M fleet of one 155 MW coal unit and two 20 MW combustion turbines. Cell $(0.3, 0.7)$ is therefore the least robust result of the table, and the necessity-side saving the most robust.

Relation to Theorem 1. At the spreads of Table S10, every one of the seven unit-specific optima coincides with the crisp optimum of some integer threshold: it belongs to the library of optima of the 73 integer thresholds of Section 5.1 of the main article, which contains only 7 distinct designs. That coincidence is not general. The \$171 M fleet that appears at $(0.3, 0.7)$ under the perturbed thermal spreads, and at $(0.5, 0.7)$ in the hydro-only variant, is the crisp optimum of no integer threshold; the exhaustive enumeration confirms both its optimality and its absence from the library. Heterogeneous spreads can therefore select a design that no single effective threshold would. The common-spread model then cannot represent the optimal design, which only the unreduced search finds.

S6.6 Policy effect of the optimism index

Section 5.2 of the main article reports the cost and the necessity reliability of the prescribed fleets. Figure S4 shows both quantities against γ. For every $\alpha < 1$ the minimum cost is non-increasing in γ, and at $\alpha = 0$ it falls from \$203 M to \$130 M; the necessity reliability of the prescribed design falls in step, from 0.95 at $(0, 0)$ to 0.00 at $(0, 1)$. All lines cross at $\gamma = 0.5$, where both panels are independent of α. Because

the spreads and the threshold are symmetric, Proposition 3 and the scope remark of Section S10 give this pattern in general: the minimum cost is non-increasing in α for $\gamma < 1/2$, which includes the necessity column, and non-decreasing in α for $\gamma > 1/2$. The necessity design costs \$203 M with $R_N = 0.9509$ and the modal design \$180 M with $R_N = 0.5056$. The Fyffe-derived instance, on which more designs are optimal, shows the same trade-off in finer steps (Section S5).

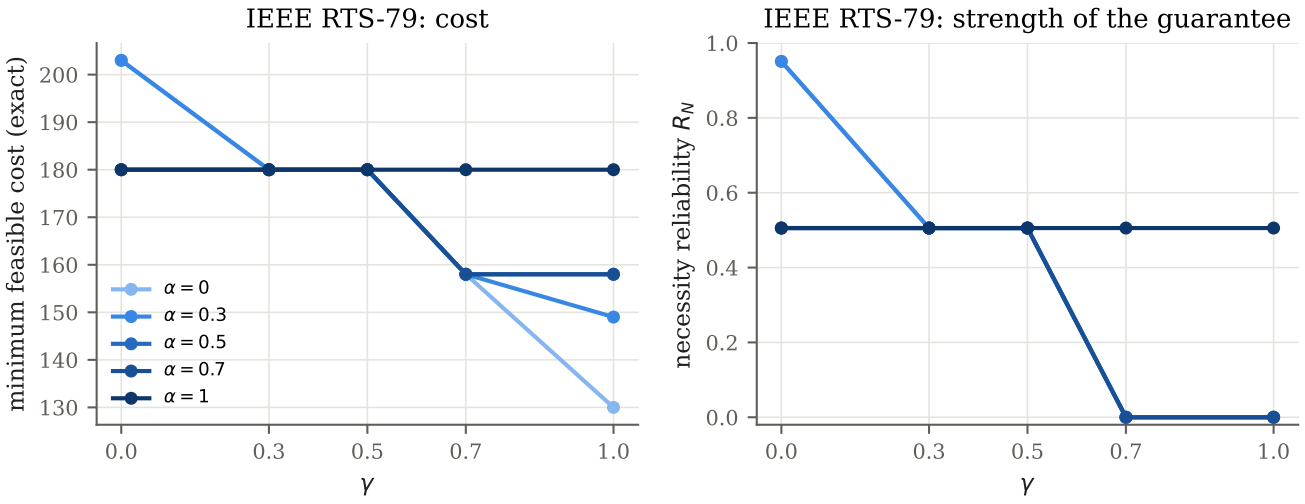


Figure S4: Policy effect of the optimism index γ on the IEEE RTS-79 fleet. Left: minimum feasible cost. Right: necessity reliability R_N of the design prescribed in each cell. The Hurwicz reliability of every prescribed design clears the target by construction and is therefore not shown; $\alpha = 0.9$ is omitted for legibility and reported in Table 5 of the main article. Several lines coincide because few designs are optimal.

S6.7 Cost–necessity frontier

The design menu (D1–D7) and a cost–necessity trade-off answer different questions. Each menu design minimizes cost subject to $R_{\text{sys}}^{\alpha,\gamma}(\mathbf{x}) \geq R_{\text{target}} = 0.95$ in its own cell; it is not selected for its necessity reliability R_N. Its crisp adequacy R_{modal}, the probability that the working modal capacity reaches 180 MW, can also differ from the target. D1–D4 install less than 180 MW (150, 155, 162 and 176 MW), so their R_{modal} is zero. Figure 2 of the main article shows both frontiers, and Table S11 lists the adequacy-constrained one up to the necessity design D7.

1. *Without the adequacy constraint* the frontier contains designs cheaper than \$180 M with positive necessity, such as the \$170 M fleet of two 100 MW oil units ($R_N = 0.4851$). Both oil units must operate to reach 180 MW, so its crisp adequacy is $R_{\text{modal}} = 0.960^2 = 0.9216 < 0.95$. Between \$162 M and \$203 M every design of this frontier except the \$180 M and \$203 M designs, which have the composition of D6 and D7, fails crisp adequacy. The two catalog fleets that cost \$171 M (1×76 MW coal $+1 \times 100$ MW oil $+1 \times 20$ MW combustion turbine, $R_N = 0.3603$; 1×155 MW coal $+2 \times 20$ MW combustion turbines, $R_N = 0.3110$) are dominated by the \$170 M fleet.

2. *With the adequacy constraint* no design cheaper than \$180 M is admissible: the cheapest design with $R_{\text{modal}} \geq 0.95$ costs \$180 M, and the frontier starts at D6 ($R_N = 0.6171$). Between D6 and D7 it contains three designs that no Hurwicz cell prescribes (Table S11): a planner bound by adequacy obtains $R_N = 0.6392$ for \$190 M, $R_N = 0.8804$ for \$194 M and $R_N = 0.9171$ for \$198 M, against 0.9509 for D7 at \$203 M.

Table S11: Adequacy-constrained cost–necessity frontier of the IEEE RTS-79 catalog: designs that minimize cost subject to $R_N(\mathbf{x}) \geq r$ and crisp adequacy $R_{\text{modal}}(\mathbf{x}) \geq 0.95$ at the modal demand of 180 MW, for every r up to the necessity design D7 (exhaustive enumeration of all 59,049 designs). The last column names the menu design (Table 4) when the frontier design belongs to it.

Cost (M$)	Fleet	MW	R_{modal}	R_N	R_Π	Menu
180	1 × 50 MW Hydro + 1 × 155 MW Coal	205	0.9504	0.6171	0.9528	D6
190	2 × 50 MW Hydro + 1 × 76 MW Coal + 1 × 12 MW Oil + 1 × 20 MW CT	208	0.9586	0.6392	0.9663	–
194	1 × 50 MW Hydro + 1 × 12 MW Oil + 1 × 155 MW Coal	217	0.9504	0.8804	0.9563	–
198	1 × 50 MW Hydro + 1 × 20 MW CT + 1 × 155 MW Coal	225	0.9504	0.9171	0.9581	–
203	3 × 50 MW Hydro + 1 × 76 MW Coal	226	0.9509	0.9509	0.9782	D7

3. *The menu and the frontier differ* because each menu design is the cheapest design at a single effective threshold T; the necessity reliability R_N does not enter that optimization and is reported for the selected design only.

A decision maker who can relax crisp adequacy may therefore adopt designs such as the \$170 M fleet, whereas one bound by it must spend at least \$180 M and can use Table S11 to price necessity levels between those of D6 and D7. The probabilistic design of Section 5.3 of the main article lies below this frontier: at \$194 M it reaches $R_N = 0.779$, against 0.880. For a capital budget, the menu and the frontier together give: \$180 M buys $R_N = 0.617$ (D6), \$203 M buys $R_N = 0.951$, and under crisp adequacy the catalog offers $R_N = 0.880$ for \$194 M and 0.917 for \$198 M.

S6.8 Capacity spread and demand forecast

This section gives the spread and demand sensitivity summarized in Section 5.5 of the main article. Asymmetry matters for generation fleets: high ambient temperature reduces the output of combustion turbines through lower air density and cooling efficiency, whereas low temperature raises it only moderately.

Symmetric spread. Figure S5(a) traces the optimal cost $Z^*(\delta)$ for δ from 0.02 to 0.20 at ten points, with the demand fixed at $\tilde{k} = (162, 180, 198)$ MW. Percentages are relative to the modal cost of \$180 M. Three postures behave differently.

1. *Necessity* $(\alpha, \gamma) = (0, 0)$. The effective threshold $T = \lceil 198/(1-\delta) \rceil$ rises from 203 MW to 248 MW, and the optimal cost from \$180 M to \$265 M (+47.2%). Under the necessity rule the cost depends only on the downside spread δ_l.

2. *Modal* $(0.5, 0.5)$. The effective threshold stays at $T = 180$ MW and the cost at \$180 M for every δ. The reason is Equation (18) of the main article: for symmetric spreads the multiplier $m(\alpha, 1/2)$ equals one, and for a symmetric threshold $k^\alpha(1/2) = k_2$, so $\kappa = k_2 = 180$ MW whatever the value of δ.

3. *Optimistic boundary* $(0, 1)$. Wider spreads let the upper capacity bounds $(1+\delta)w_i$ meet the requirement. At $\delta = 0.20$ the effective threshold falls to $T = \lceil 162/(1+\delta) \rceil = 135$ MW and the cost to \$127 M (−29.4%). The prescribed design, however, has zero necessity of meeting demand ($R_N = 0$, $R_\Pi = 0.0751$).

Asymmetric spreads. The markers in Figure S5(a) give the necessity-cell cost under four stylized seasonal regimes applied to all units: summer, $(\delta_l, \delta_r) = (0.15, 0.05)$, \$231 M at $T = 233$ MW; extreme

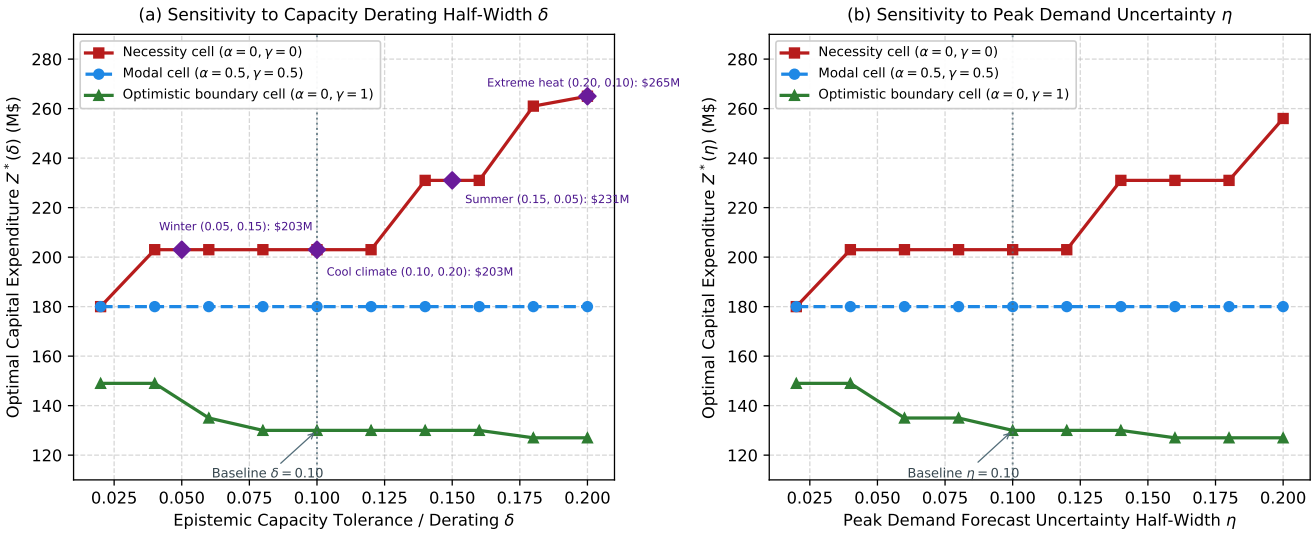


Figure S5: Sensitivity of optimal capital expenditure to epistemic capacity spread δ and peak demand forecast uncertainty η on the IEEE RTS-79 generation fleet. (a) Sensitivity over the symmetric capacity spread $\delta \in [0.02, 0.20]$; diamonds give the necessity-cell cost under four asymmetric regimes (δ_l, δ_r), plotted at $\delta = \delta_l$. (b) Sensitivity to peak demand forecast uncertainty half-width $\eta \in [0.02, 0.20]$.

heat, $(0.20, 0.10)$, \$265 M at $T = 248$ MW; winter, $(0.05, 0.15)$, \$203 M at $T = 209$ MW; and cool climate, $(0.10, 0.20)$, \$203 M at $T = 220$ MW, as for the symmetric 10% spread, because the upside does not enter the necessity rule. Under every asymmetric regime the modal cell $(0.5, 0.5)$ moves as well, because $m(0.5, 0.5) = 1 + (\delta_r - \delta_l)/4 \neq 1$: its threshold is $T = 185$ MW (\$180 M) for summer and extreme heat and $T = 176$ MW (\$158 M) for winter and cool climate. Asymmetric spreads need no additional computation: Theorem 1 allows $\delta_l \neq \delta_r$, so every regime is solved through Equation (18) with the same certificate.

Demand uncertainty. Figure S5(b) varies the half-width η of the demand forecast $\tilde{k} = (180(1 - \eta), 180, 180(1 + \eta))$ from 0.02 to 0.20. Under the necessity rule the cost rises from \$180 M to \$256 M; in the optimistic boundary cell it falls to \$127 M.

S7 Model Baselines: Details

Section 4.2 of the main article defines the model baselines and Section 5.3 reports their main results. Table S12 gives the decision-level baselines in full, including the share of the square on which the baseline design is also optimal and the area-weighted mean cost excess, and Table S13 the probabilistic counterpart under both densities.

On the RTS-79 fleet the three decision-level baselines fail on the same region, on which the dual threshold exceeds 200 MW and every baseline prescribes the four-hydro design D5. Elsewhere the crisp design is D5 at every posture and is optimal on the 61.1% of the square on which D5 is optimal; the one-sided models track part of the variation of the threshold and are optimal on about three quarters of the square. On the derived instances, whose libraries contain 29 and 20 designs, the one-sided models fail on a third of the square and the crisp design on half.

The probabilistic designs meet the target under the declared rule only up to an effective threshold of 200 MW on the RTS-79 fleet and 41 on the Pharmaceutical-derived instance; the random threshold exceeds these values with probability 0.034 and 0.373 under triangular densities. Each probabilistic design is optimal at no posture of the square. The choice of density changes the design on both instances, while its threshold coverage and its failure and higher-cost shares over the square stay the same; the largest cost excess changes slightly.

Table S12: Decision-level model baselines in full (shares of the part of the policy square on which some design meets the target). "Optimal": the baseline design is also optimal under the declared rule; "Mean excess": area-weighted relative cost excess over the part on which the baseline design meets the target, including the part on which it is optimal.

Instance	Baseline	Fails (%)	Costlier (%)	Optimal (%)	Max. excess (%)	Mean excess (%)
RTS-79	Crisp	6.8	32.1	61.1	38.5	5.5
	Fuzzy capacities only	6.8	17.1	76.1	21.5	2.2
	Fuzzy demand only	6.8	16.7	76.5	21.5	2.1
Fyffe-derived	Crisp	50.0	37.6	12.4	64.3	12.0
	Fuzzy capacities only	37.7	37.6	24.7	42.9	6.6
	Fuzzy demand only	33.0	27.7	39.3	21.4	2.8
Pharma-derived	Crisp	49.9	33.4	16.7	33.7	5.5
	Fuzzy capacities only	33.5	33.4	33.1	25.0	3.0
	Fuzzy demand only	30.9	30.3	38.8	16.3	1.9

Table S13: Probabilistic counterpart: optimal design under triangular and uniform densities on the supports of the fuzzy data, the largest effective threshold at which it meets the target, the probability that the random threshold exceeds it, and its evaluation over the policy square under the declared rule. Pharmaceutical designs are given as option vectors.

Instance	Density	Design	Cost	R_{prob}	T met	$\Pr\{\hat{T} > \cdot\}$	Fails (%)	Costlier (%)	Max. exc. (%)
RTS-79	triangular	4×50 MW hydro + 1×12 MW oil	194	0.9605	200	0.034	6.8	93.2	49.2
	uniform	4×50 MW hydro + 1×20 MW CT	198	0.9587	200	0.112	6.8	93.2	52.3
Pharma-derived	triangular	(0, 2, 1, 1, 1, 2, 0, 2, 1, 2)	133	0.8513	41	0.373	33.5	66.5	44.6
	uniform	(0, 2, 1, 1, 2, 2, 0, 2, 1, 2)	135	0.8541	41	0.417	33.5	66.5	46.7

S8 Replication on the Literature-Derived Instances

Section 5.1 of the main article reports the complete parametric solution of the IEEE RTS-79 fleet and what a 30-cell grid misses. Table S14 repeats the comparison on the two derived instances of Sections S4 and S5.

Table S14: The same quantities for the case study and the two literature-derived instances (30-cell grid and complete parametric solution of Theorem 1). A design "at no grid cell" is optimal on some region of the policy square but at none of the 30 grid cells; the last four rows describe the region on which such designs are optimal: its area, the share of it on which the design of the nearest grid cell (closest α and γ) does not meet the reliability target, the share of the square on which no grid design meets the target, and the largest cost excess over the optimum of the cheapest grid design that meets it. Costs are in the units of each catalog (M$ for RTS-79).

	IEEE RTS-79	Fyffe-derived	Pharmaceutical-derived
Design space	$3^{10} = 59{,}049$	$4^{14} = 2.68 \times 10^8$	$3^{10} = 59{,}049$
Range of $\kappa(\alpha, \gamma)$	[147.27, 220.00]	[36.36, 66.67]	[30.91, 51.11]
Integer thresholds (with a feasible design)	73 (73)	31 (31)	22 (20)
Distinct optimal designs on the square	7	29	20
Distinct optimal designs at the grid cells	5	15	14
Optimal designs at no grid cell	2	14	6
Minimum cost on the grid	130–203	14–36	92–158
Grid cells with a feasible design	30/30	30/30	29/30
Area optimal at no grid cell (% of square)	4.8	22.5	8.5
nearest cell misses target (% of that area)	53	51	65
no grid design meets target (% of square)	0.0	0.0	1.6
max. cost excess of grid design (%)	12.8	13.3	8.4

Both findings recur. First, the policy square splits into finitely many regions: 73, 31 and 22 integer thresholds yield 7, 29 and 20 distinct optimal designs, so the complete parametric solution needs a number of integer problems of the same order as the 30-cell grid. Second, the grid misses part of the solution in every instance, and the part it misses matters. 2 of 7, 14 of 29 and 6 of 20 optimal designs are optimal at no grid cell, and their regions cover 4.8%, 22.5% and 8.5% of the square. On 53%, 51% and 65% of these regions the design of the nearest grid cell does not meet the reliability target, and the cheapest grid design that does meet it costs up to 12.8%, 13.3% and 8.4% more than the optimum. On 1.6% of the square of the Pharmaceutical-derived instance no grid design meets the target, although a design that meets it exists. These figures evaluate the archived designs at each effective threshold and integrate the regions of Corollary 2; no further optimization is involved. A finer grid would reduce them but cannot rule such cases out, whereas the parametric solution covers every point of the square.

The effect of the posture also recurs on the derived instances, whose data are also symmetric: the minimum cost spans 14–36 and 92–158 in the units of their catalogs, is highest on the necessity column $\gamma = 0$ and varies with α as on the fleet (Table S14; Sections S4 and S5). The Pharmaceutical-derived instance adds a case the fleet does not show: in the necessity cell $(0, 0)$ no design meets the reliability target, and the two largest integer thresholds have no feasible design.

S9 Computational Performance

S9.1 Scaling family

We solved a family of synthetic catalogs with three choices per subsystem at $n \in \{8, 10, \ldots, 20\}$ under three demand profiles (*screening*, *design* and *stringent*, with loose, moderate and tight thresholds), 21 instances in total, with the reduction of Theorem 1, a compiled recursion and a 600 s budget per instance.

All 21 completed with a certificate.

Table S15: Certified exact solution of the synthetic scaling family (3 choices per subsystem, design profile shown; T is the effective integer threshold of Theorem 1). "Leaves" counts complete designs actually evaluated. The instances $n = 8$ and 10 (solved in at most 0.012 s) are omitted.

n	Search space 3^n	T	Time (s)	Nodes	Leaves	Optimal cost
12	5.31×10^5	60	0.03	5.66×10^4	7	63
14	4.78×10^6	66	0.27	6.09×10^5	35	77
16	4.30×10^7	78	6.20	1.49×10^7	29	99
18	3.87×10^8	81	30.41	6.23×10^7	47	111
20	3.49×10^9	100	459.24	8.35×10^8	89	117

Table S15 reports the design profile, the hardest of the three. Two quantities describe the pruning. The node count measures the work of the search, including the internal nodes of partial designs: at $n = 20$ the solver expands 8.35×10^8 nodes of a full ternary tree of $(3^{21} - 1)/2 = 5.23 \times 10^9$ nodes, or 15.96%. This count is comparable with the size of the tree, not with 3^{20}, which counts complete designs. The leaf count is the number of complete candidate designs reaching depth n whose exact reliability is evaluated: 89 of 3.49×10^9. The admissible bounds thus discard almost every complete design at interior levels before constructing it.

S9.2 Ablation

Progress from $n = 14$ to $n = 20$ combines an algorithmic change and a larger budget. Table S16 gives the frontier of each regime.

Table S16: Largest instance certified under each regime (design profile, 12 s iso-budget unless stated).

Regime	Configuration	Frontier
1	Interpreted backend, unreduced threshold, 12 s	$n = 12$ ($n = 14$ times out)
2	Compiled backend, reduced threshold, 12 s	$n = 16$ (6.20 s; $n = 18$ times out)
3	Compiled backend, reduced threshold, 600 s	$n = 20$ (459.24 s; 0 censored)

Regimes 1 and 2 share the budget and differ only in formulation and implementation: the certified frontier moves from $n = 12$ to $n = 16$. To separate the reduction from the compiler, we re-solved the $n = 14$ design instance with the same compiled backend but without the reduction, that is, against a threshold obtained by scaling the continuous fuzzy endpoints into integers (scale factor 10^4, giving an unreduced threshold of 16,767 against $T = 66$ for the reduced model). It required 60.69 s, against 0.268 s for the reduced formulation, a factor of 226.8. The factor is close to the ratio of the thresholds ($16{,}767/66 \approx 254$), as expected, because the convolution table is proportional to the integer threshold. The gain from Theorem 1 therefore comes from replacing a finely scaled threshold by the modal integer threshold, without any discretization of the state. Regime 3 shows that a larger budget extends certification to $n = 20$.

The reduction does not change the solutions. Across all 12 instances up to $n = 14$, the reduced and unreduced formulations agreed on cost, reliability and design. Above $n = 14$ the unreduced solve exceeds the budget and returns no verdict, so no agreement is claimed there. Neither the bounds nor the reduction removes the exponential worst case; on the instances studied, certified optima are obtained within the reported budgets, and every cell of the RTS-79 fleet solves in at most 0.01 s.

S9.3 Genetic-algorithm baseline

Because metaheuristics are the usual method in this literature, we compared the exact solver with a genetic algorithm (population $\max(60, 12n)$, ternary tournament selection, uniform crossover, per-gene

mutation with rate $1.5/n$, elitism of two individuals, and feasibility-first lexicographic comparison) across 17 instances (Table S17): the Pharmaceutical-derived and Fyffe-derived instances, and the 15 non-trivial synthetic scaling instances ($n \in \{12, 14, 16, 18, 20\}$ under all three demand profiles; the small instances $n \in \{8, 10\}$ solve in at most 0.012 s and were excluded as trivial for heuristic benchmarking).

A fair comparison requires matched computational effort. A heuristic that stops at its own stagnation rule after a few seconds, while the exact solver runs for minutes, uses only a fraction of the budget. We therefore made the heuristic restart on stagnation so that it uses its whole budget, and ran two protocols. Protocol A gives the heuristic the same wall-clock time the exact solver used on that instance, with a minimum of 5 s. Protocol B gives it 1,000,000 objective evaluations on the two instances where the outcome differed; because the heuristic is interpreted and the exact solver compiled, wall-clock matching favors the exact solver, whereas an evaluation budget does not. In both protocols the heuristic's incumbent is then refined by an equal-cost local search (single-subsystem substitutions of identical cost and pairwise swaps whose cost changes cancel), the refinement a practitioner would apply when the secondary objective is reliability.

After this refinement the heuristic matched the certified optimum on both objectives in all five seeds on 16 of the 17 instances. The exception is the largest instance, $n = 20$ under the design profile, where all five seeds reached the minimum cost of 117 but one seed under Protocol A, and two under Protocol B, settled on a design whose reliability is 0.9035 against the certified 0.9112, a shortfall of 7.7×10^{-3}. The refinement did not close the gap: it terminated after 16 to 23 evaluations without improvement, because the better design cannot be reached from the incumbent through cost-preserving moves. At $n = 14$ under the same profile, one seed missed the secondary optimum before refinement, and the refinement recovered it.

Table S17: Genetic-algorithm baseline against the certified optimum, five seeds per instance. "Cost" counts seeds reaching the minimum cost; "Secondary" counts those also reaching the maximum-reliability design at that cost.

Instance	Protocol	Budget	Cost	Secondary	Shortfall
15 other instances	A	matched wall-clock	5/5	5/5	—
$n = 14$ design	A	5.0 s	5/5	$4/5 \rightarrow 5/5$ after refinement	0
$n = 14$ design	B	10^6 evaluations	5/5	5/5	0
$n = 20$ design	A	365.2 s[a], 2.50×10^6 eval.	5/5	4/5	7.7×10^{-3}
$n = 20$ design	B	10^6 evaluations, 146 s	5/5	3/5	7.7×10^{-3}

[a] The exact runtime for $n = 20$ is recorded as 365.2 s under Protocol A and 459.2 s under the scaling campaign; the difference reflects separate runs with different profiling overhead on the host.

Without a certificate, nothing in the heuristic's output indicates that a more reliable design exists at the same cost.

S10 Proofs

This section gives the proofs of the results stated in Sections 2 and 3 of the main article.

Proof of Proposition 1. Take the necessity branch with $a_2 > b_2$. Writing out $N \geq \lambda$ using the closed form gives $a_1 + (1 - \lambda)(a_2 - a_1) \geq b_3 - (1 - \lambda)(b_3 - b_2)$, whose two sides are precisely the lower α-cut endpoint of $\tilde{A}$ and the upper α-cut endpoint of $\tilde{B}$ evaluated at $\alpha = 1 - \lambda$; by Equations (2) and (3) of the main article these are $W_S^{\alpha}(0)$ and $k^{\alpha}(0)$. When $a_2 \leq b_2$ the necessity is zero, so for $\lambda > 0$ the constraint $N \geq \lambda$ fails, and under the non-degenerate-spread assumption the pessimistic comparison fails as well. The restriction to $\lambda > 0$ is needed: at $\lambda = 0$ the constraint $N \geq 0$ holds identically while the cell $(1, 0)$ still demands $a_2 \geq b_2$, so the two are not equivalent there, and the possibility branch behaves in the same way at $\lambda = 0$. The possibility branch is symmetric, exchanging the roles of the endpoints and reading $\alpha = \lambda$ at $\gamma = 1$. □

Proof of Corollary 1. For triangular numbers $N(\tilde{A} \geq \tilde{B}) > 0$ requires $a_2 > b_2$, and then $\Pi(\tilde{A} \geq \tilde{B}) = 1$. Let $\lambda > 1/2$. If $N = 0$, then $\mathrm{Cr} = \Pi/2 \leq 1/2 < \lambda$; if $N > 0$, then $\mathrm{Cr} = (1 + N)/2$. Hence $\mathrm{Cr} \geq \lambda$ if and only if $N \geq 2\lambda - 1$. Let $\lambda \leq 1/2$. If $N > 0$, then $\mathrm{Cr} \geq 1/2 \geq \lambda$ and $\Pi = 1 \geq 2\lambda$; if $N = 0$, then $\mathrm{Cr} = \Pi/2$. Hence $\mathrm{Cr} \geq \lambda$ if and only if $\Pi \geq 2\lambda$. Since $2\lambda - 1 \in (0, 1]$ in the first case and $2\lambda \in (0, 1]$ in the second, Proposition 1 maps the two conditions to the cells $(2 - 2\lambda, 0)$ and $(2\lambda, 1)$. □

The expected credibility reliability, the probability-weighted average of Cr over the subsets, is not a cell: like R_N and R_Π, it averages over the cut level. At $\lambda = 1/2$ the cell is $(1, 1)$, the modal comparison $W_S \geq k_2$.

The central column. With symmetric spreads and a symmetric threshold the central column $\gamma = 1/2$ gives the modal comparison $W_S \geq k_2$, and hence the credibility constraint at $\lambda = 1/2$, at every α. With an asymmetric threshold it does not: for a crisp capacity $W_S = 42$ and $\tilde{k} = (30, 40, 60)$ the cell $(0, 1/2)$ requires $W_S \geq k^0(1/2) = 45$, whereas $\mathrm{Cr} \geq 1/2$ requires only $W_S \geq 40$.

Proof of Proposition 2. Fix a subset S and design $\mathbf{x}$. By Proposition 1, for every $\alpha \in [0, 1)$ the indicator of the cell $(\alpha, 0)$ accepting S equals the indicator of $N(\tilde{W}_S \geq \tilde{k}) \geq 1 - \alpha$; the single point $\alpha = 1$ does not affect the integral. Hence $\int_0^1 \mathbf{1}\{S \text{ accepted at } (\alpha, 0)\}\, d\alpha = N(\tilde{W}_S \geq \tilde{k})$, which is the layer-cake representation of a number in $[0, 1]$. Multiplying by $\Pr(S \mid \mathbf{x})$ and summing over all subsets (both operations are linear and the sum is finite) yields the first identity; the second follows symmetrically at $\gamma = 1$. □

Proof of Proposition 3 and scope of the monotonicity. Under the stated derivative signs, increasing α cannot improve the success inequality $W_S^\alpha(\gamma) \geq k^\alpha(\gamma)$ for any fixed subset S. Hence, if a subset is feasible at α_2, it is also feasible at any $\alpha_1 < \alpha_2$, and since $R_{\mathrm{sys}}^{\alpha,\gamma}$ sums nonnegative subset probabilities over these feasible sets, the ordering follows. □ The sign conditions are sufficient, not universal. For a component TFN with left and right spreads $\Delta_i^- = w_i - w_{i,L}^0$ and $\Delta_i^+ = w_{i,U}^0 - w_i$, and a threshold with $\Delta_k^- = k_2 - k_1$ and $\Delta_k^+ = k_3 - k_2$,

$$\frac{d}{d\alpha} w_i^\alpha(\gamma) = (1 - \gamma)\Delta_i^- - \gamma\Delta_i^+, \qquad \frac{d}{d\alpha} k^\alpha(\gamma) = \gamma\Delta_k^- - (1 - \gamma)\Delta_k^+, \tag{S.5}$$

so, for non-degenerate spreads, the conditions hold if and only if $\gamma \geq \Delta_i^-/(\Delta_i^- + \Delta_i^+)$ for every installed component and $\gamma \geq \Delta_k^+/(\Delta_k^- + \Delta_k^+)$. For symmetric data both reduce to $\gamma \geq 1/2$; a left-skewed TFN moves the critical index towards 1 (for $\Delta_i^- = 10$, $\Delta_i^+ = 1$ the capacity derivative is positive for every $\gamma < 10/11$). For symmetric data and $\gamma < 1/2$ both signs are reversed and the same argument gives the reversed ordering $R_{\mathrm{sys}}^{0,\gamma} \leq R_{\mathrm{sys}}^{\alpha,\gamma} \leq R_{\mathrm{sys}}^{1,\gamma}$; at $\gamma = 1/2$ both derivatives vanish. For general asymmetric data neither ordering need hold, and the feasible-subset family need not be nested.

Proof of Corollary 2. Equating $\kappa(\alpha, \gamma) = T$ in Equation (18) gives $k^\alpha(\gamma) = T \cdot m(\alpha, \gamma)$. Substituting $k^\alpha(\gamma) = k_U^\alpha - \gamma(k_U^\alpha - k_L^\alpha)$ and $m(\alpha, \gamma) = 1 - \delta_l(1 - \alpha) + \gamma(\delta_l + \delta_r)(1 - \alpha)$ yields an equation linear in γ, whose solution is Equation (19). At $\alpha = 1$ all triangular numbers collapse to their modal values, $k_U^1 = k_L^1 = k_2$ and $m(1, \gamma) = 1$, so $\kappa(1, \gamma) = k_2$ for every γ and the denominator vanishes. □

Proof of Theorem 2. (i) Capacities are non-negative, so the capacity of any subset of a completion is at most the capacity of all its units, and $W_{\mathrm{part}} + w_{i,j} + \sum_{l>i} w_l^{\max}$ bounds the latter from above. If this bound is below $k^\alpha(\gamma)$, no subset reaches the threshold. (ii) Any feasible completion must supply at least q_{next} further units of capacity, so by the definition of Z_i^{suf} its additional cost is at least $Z_i^{\mathrm{suf}}(q_{\mathrm{next}})$; since costs are non-negative, its total cost is at least $Z_{\mathrm{next}} + Z_i^{\mathrm{suf}}(q_{\mathrm{next}})$. If this exceeds Z^* strictly, every completion is strictly more expensive than the incumbent and is dominated at the primary level of the ordering. A completion of equal cost may still improve the secondary objective, and discarding it would forfeit lexicographic optimality. □

Proof of Theorem 3. The structure is coherent, so the static reliability is non-decreasing in every component reliability and every capacity. Each virtual option dominates every catalog option of its subsystem in both, so the convolved probability of reaching the threshold $k^\alpha(\gamma)$ is at least the reliability of any completion. □

Proof of Theorem 4. The search tree has $\prod_{i=1}^{n} |M_i|$ leaves, a finite number. In Phase 1 the solver prunes only subtrees whose cost lower bound strictly exceeds the incumbent (Theorem 2(ii)) or whose reliability upper bound falls strictly below R_{target} (Theorem 2(i) and Theorem 3). Since these bounds are admissible, no feasible completion with cost strictly below the incumbent is ever discarded, so Phase 1 returns the minimum feasible cost Z^*, or infeasibility if no leaf meets the target. Any feasible cost is a valid incumbent, so the greedy seed cannot change the certified optimum. In Phase 2 the solver restricts the search to the designs of cost Z^* by pruning any partial assignment with $Z_{\text{next}} + Z_i^{\text{suf}}(q_{\text{next}}) > Z^*$. All examined leaves have cost Z^*, and the algorithm records the design $\mathbf{x}^*$ with the largest reliability R^*. Since the finite search space is inspected exhaustively up to admissible pruning, $\mathbf{x}^*$ is lexicographically optimal. □

S11 Notation

Table S18 lists the notation of the main article.

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Table S18: Nomenclature.

Indices and sets	
n	Number of subsystems (components)
i	Subsystem index, $i \in \{1, \dots, n\}$
j	Catalog design alternative index, $j \in M_i$ ($j = 0$: no unit installed)
M_i	Discrete catalog of available design alternatives for subsystem i
$I(\mathbf{x})$	Installed subsystems of design $\mathbf{x}$, $I(\mathbf{x}) = \{i : x_i \neq 0\}$
S	Subset of operational subsystems, $S \subseteq I(\mathbf{x})$
$\mathbf{s}$	Component state vector, $s_i = 1$ if component i works and 0 otherwise
Decision variables and physical parameters	
x_i, $\mathbf{x}$	Selected alternative for subsystem i; design vector $\mathbf{x} = (x_1, \dots, x_n)$
$c_{i,j}, c_i(x_i)$	Capital cost of choice j in subsystem i
$p_{i,j}$	Reliability of choice j in subsystem i
$\tilde{w}_i$	Triangular fuzzy capacity of component i, $(w^0_{i,L}, w_i, w^0_{i,U})$
W_S, $\tilde{W}_S$	Modal and fuzzy aggregate capacity of subset S, $\tilde{W}_S = \sum_{i \in S} \tilde{w}_i$
δ_l, δ_r	Common proportional left and right capacity spreads (δ when equal)
$\tilde{k}$	Triangular fuzzy demand threshold, (k_1, k_2, k_3)
η	Relative half-width of the demand forecast in the sensitivity study
R_{target}	Prescribed minimum system reliability
Possibilistic and operational metrics	
α, γ	Cut level and Hurwicz optimism index, both in $[0, 1]$
$w^\alpha_{i,L}, w^\alpha_{i,U}$; k^α_L, k^α_U	Lower and upper α-cut endpoints of $\tilde{w}_i$ and $\tilde{k}$
$k^\alpha(\gamma)$, $w^\alpha_i(\gamma)$	Hurwicz values of the threshold and of capacity i
$m(\alpha, \gamma)$	Capacity multiplier under common proportional spreads
$\kappa(\alpha, \gamma)$, T	Effective threshold $k^\alpha(\gamma)/m(\alpha, \gamma)$ and its integer value $T = \lceil \kappa \rceil$
$\gamma_T(\alpha)$	Boundary of the region $\{\lceil \kappa \rceil = T\}$ in the policy square
λ	Confidence level of a possibilistic chance constraint, $\lambda \in (0, 1]$
Π, N, Cr	Possibility, necessity and credibility measures
$R^{\alpha,\gamma}_{\text{sys}}(\mathbf{x})$	Static (Hurwicz) reliability of design $\mathbf{x}$ under rule (α, γ)
$R_N(\mathbf{x}), R_\Pi(\mathbf{x})$	Expected necessity and possibility reliability of design $\mathbf{x}$
$R_{\text{modal}}(\mathbf{x})$, $R(\mathbf{x}, T)$	Crisp reliability at the modal demand k_2 and at the integer threshold T
D, Θ, $\hat{T}$	Random demand, common derating factor and random threshold of the probabilistic counterpart
$R_{\text{prob}}(\mathbf{x})$	Reliability under the probabilistic counterpart
Optimization and search	
$Z(\mathbf{x})$, Z^*	Total capital cost; minimum feasible cost (incumbent during the search)
$\mathcal{X}^*$	Set of minimum-cost feasible designs
$\bar{Z}$	Capital budget
r	Required expected necessity reliability on the cost–necessity frontier
$Z^{\text{suf}}_i(q)$	Minimum cost of subsystems $i+1, \dots, n$ that supply at least q further capacity units
$W^{\max}_i$	Largest capacity that subsystems $i+1, \dots, n$ can add
ϵ	Numerical tolerance on reliability comparisons, $\epsilon = 10^{-12}$
R_{ub}	Optimistic upper bound on the reliability of any completion of a partial design
Abbreviations	
FOR	Forced outage rate of a generating unit
IEEE RTS-79	IEEE Reliability Test System (1979)
MTTF, MTTR	Mean time to failure, mean time to repair
TFN	Triangular fuzzy number
Wk/n:G	Weighted k-out-of-n:G (system)